Small-Model Alignment
SFT, DPO and Persona LoRA

Advanced Generative AI - PA2
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Repository: github.com/namranadeem-wq/25280097-Advanced-Gen-AI-PA2

1. Outcome and experimental design

This study trains Qwen/Qwen2.5-0.5B with response-only supervised fine-tuning, then runs two independent full-parameter DPO experiments from the same best SFT checkpoint. A separate TRL/LoRA experiment steers HuggingFaceTB/SmolLM2-360M-Instruct toward a Ramsay-inspired persona. Training used Kaggle Tesla T4 GPUs; the full DPO policy and frozen reference were placed on separate GPUs.

Higher validation preference accuracy did not reliably improve generated answers. On anonymous Gemini 3.6 Flash comparisons, UltraFeedback DPO lost more often than it won against SFT. HH DPO was close to SFT, while the persona adapter mostly produced ties against the original instruction model. These are model-judge preferences on fixed subsets, not evidence of universal quality or safety.

Comparison	Pairs	Wins / losses / ties	Win rate*
UF DPO vs SFT	300	103 / 138 / 59	42.74%
HH DPO vs SFT	300	95 / 102 / 103	48.22%
Persona LoRA vs original	22	3 / 3 / 16	50.00%

*Wins divided by wins + losses. The candidate is the DPO/LoRA model. All 622 pairs received valid judgments; unknown and final unresolved counts were zero.

2. Data and fixed evaluation sets

SFT used 30,000 training and 1,000 validation instruction-response examples from HuggingFaceH4/ultrachat_200k (train_sft and test_sft), extracting the first user/assistant pair. HH DPO used 15,000/1,000 single-turn pairs from Anthropic/hh-rlhf (train/test). UltraFeedback DPO used 15,000/1,000 pairs from HuggingFaceH4/ultrafeedback_binarized (train_prefs/test_prefs). Preparation used seed 42, excluded duplicate prompts across each train/validation split and exact evaluation-prompt matches, and filtered empty responses and overlong prompts. These preparation choices are recorded in notebook history; preparation scripts were not included in the evidence archive.

General evaluation used a fixed 300-prompt AlpacaEval subset with seed 42. Safety/helpfulness evaluation used eval_sets/hh_rlhf_eval_300_seed42.json, the supplied synthetic 300-prompt set. Persona data contained 220 pairs: 198 training and 22 validation examples after a seeded shuffle. The same 22 prompts were used for checkpoint selection and generation evaluation, so persona results are validation-set analysis, not an independent test.

3. Implementation and actual configuration

alignment_core.py tokenizes the prompt and response separately without special tokens, appends response EOS, truncates the concatenated sequence, and right-pads batches. Prompt and padding labels are -100; attention remains enabled for real prompt tokens. Shifted logits at position t score labels at t+1. FP32 cross-entropy supplies response-token log probabilities, summed per row. SFT uses a token-mean reduction, consistently normalized across each accumulation group.

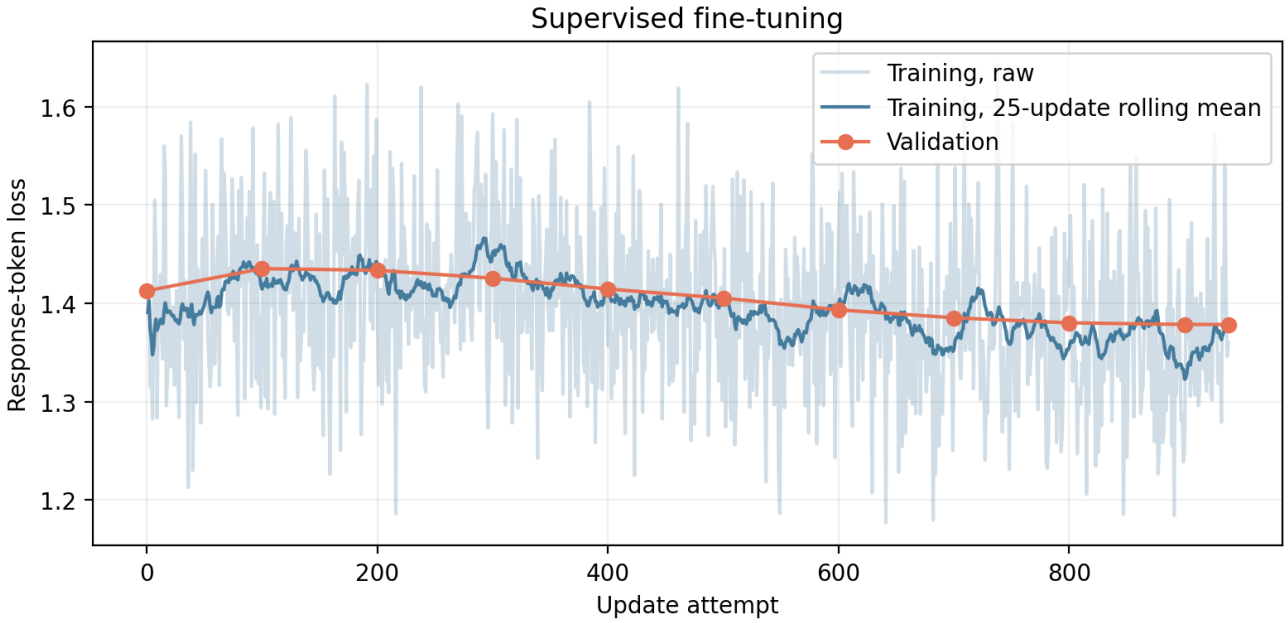
For each DPO pair, the code evaluates chosen and rejected response sums for policy and reference. With delta = (log p_policy(chosen) - log p_policy(rejected)) - (log p_ref(chosen) - log p_ref(rejected)), the objective is mean[-log sigmoid(beta * delta)]. Reference evaluation runs under no_grad. Reward margin is beta * delta; reward accuracy is the fraction with margin strictly greater than zero. The root submission_adapter.py exposes the required thin wrappers. The recorded public smoke-test result was 7 passed; passing those checks does not guarantee private-test correctness.

Setting	SFT	HH / UF DPO	Persona LoRA-DPO
Strategy	Full parameters	Full policy; frozen SFT reference	1,638,400 trainable / 363,459,520 total (~0.45%)
Train / validation	30,000 / 1,000	15,000 / 1,000 each	198 / 22
Optimizer; LR	AdamW; 2e-5	AdamW; fixed 1e-6	AdamW; 5e-5
Schedule / warmup	Cosine / 29 steps	None / none	Cosine / 10%
Weight decay / clip	0.1 / 1.0	0 / 1.0	0 / 1.0
Batch / accumulation / effective	2 / 16 / 32	1 / 32 / 32	2 / 4 / 8
Length / epochs / seed	512 / 1 / 42	512 / 1 / 42	512 / 3 / 42
Precision	FP32 weights; FP16 AMP	FP32 policy + FP16 AMP; FP16 reference	FP32 base loading; FP16 AMP
DPO beta / reference	Not applicable	0.1 / outputs/sft/best	0.1 / original instruction model, adapters disabled
Selection	Lowest validation token loss	Highest validation reward accuracy	Highest validation reward accuracy
Updates	938 successful	469 attempts; 466 successful each	75 steps; selected step 25

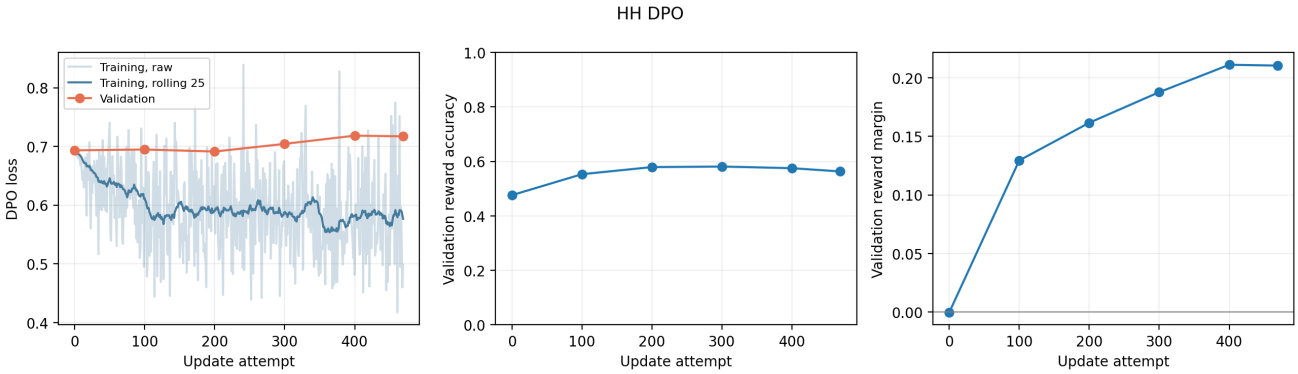
All runs used gradient checkpointing. DPO dropout was disabled. LoRA used rank 8, alpha 16 and q_proj/k_proj/v_proj/o_proj targets. Although configured lora_dropout was 0.05, TRL disable_dropout=True made its effective value zero. The report therefore records zero effective dropout rather than treating 0.05 as active regularization.

Prompt template: ### Instruction: followed by the instruction, two newlines, then ### Response:. This same template was used for Qwen training and evaluation. Persona inputs used the original model chat template with a user message and assistant generation prefix.

4. Training behavior and checkpoint selection



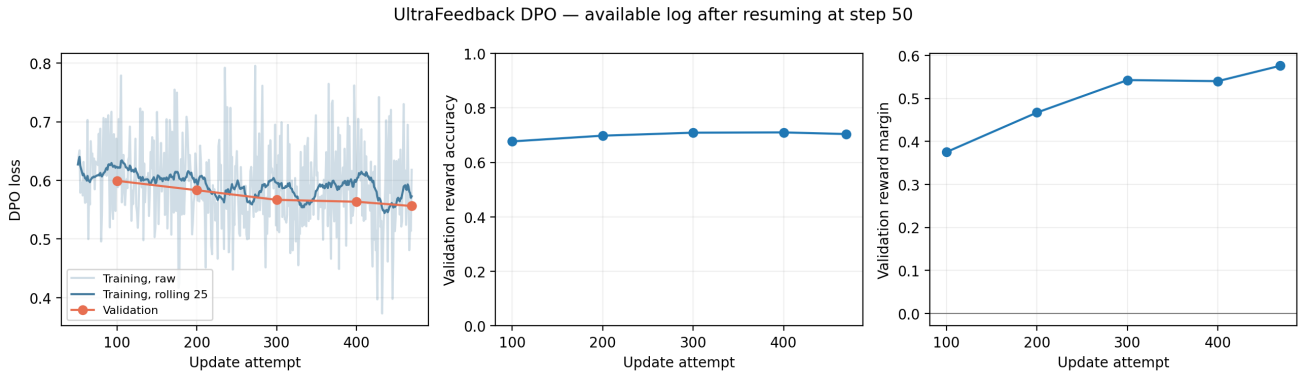
SFT: validation token loss rose from 1.4127 at initialization to 1.4353 at step 100, then declined to 1.37846 at step 938. Best and final weights coincide. The final loss is about 2.4% below the initial value. No optimizer updates were skipped. The noisy raw training curve is shown with a 25-update rolling mean; smoothing is descriptive and is not used for checkpoint selection.



HH DPO: validation reward accuracy peaked at 58.1% at step 300, selecting that checkpoint. Final accuracy fell to 56.3%, while validation loss increased to 0.7172 and mean margin rose to 0.2104. Increasing average margin with worsening loss and accuracy is consistent with uneven confidence across pairs, rather than uniform improvement. Later optimization did not improve the chosen selection metric.

The initial HH reward accuracy was 47.6% with mean margin -0.000409. This is numerical asymmetry, not learned preference: the policy and reference used different weight precisions. Exactly identical computations would give zero margins and zero strict reward accuracy. Three of 469 update attempts were skipped by gradient scaling; the logs retain their non-finite gradient norms.

4. Training behavior (continued)



UltraFeedback DPO: validation loss declined from 0.5993 at step 100 to 0.5563 at step 469. Accuracy peaked at 71.0% at step 400 and ended at 70.4%; step 400 is the evaluation checkpoint. Mean margin grew from 0.3754 to 0.5759 over these validation points. These improvements establish learning on preference pairs, but the generation comparison below shows they are not sufficient evidence of better instruction-following.

The accepted run resumed from a downloaded step-50 state, restoring model, optimizer, scaler, data permutation, cursor, and torch/CUDA RNG state. Its durable metrics begin at step 51; the plot does not fabricate the missing first 50 steps. The run used a fixed torch.randperm with seed 42. It completed 466 successful updates out of 469 attempts. Earlier interrupted attempts are not included in the accepted-run totals.

Persona: rapid preference learning, weak generation change

TRL trained for 75 steps over three epochs. Validation reward accuracy was 100% at steps 25, 50 and 75, so the first maximum, step 25, was retained. Its validation loss was 0.5957 and reward margin 0.2063. Later checkpoints reduced loss to 0.4633 and raised margin to 0.5352 but did not improve accuracy. Re-evaluating the restored best adapter at the end reproduces the step-25 metrics; the final log row is not a second independent step-75 checkpoint measurement.

Parameter choices and deviations

The full SFT/DPO configurations follow the manual recipe: context 512, effective batch 32, SFT LR 2e-5 with cosine decay, and DPO beta 0.1 with fixed LR 1e-6. AdamW is one of the permitted DPO optimizer choices. Splitting policy/reference across two T4 GPUs and using a half-precision frozen reference address memory constraints, with the numerical asymmetry noted above.

Persona hyperparameters were student choices rather than a prescribed optimum. Rank 8 with alpha 16 provides modest capacity (~0.45% trainable parameters). Attention projections offer a targeted way to change response behavior while retaining the base. LR 5e-5, effective batch 8 and three epochs provide a short experiment on 198 pairs. These are reasonable starting choices, not a tuned optimum: no rank/LR ablation was conducted. Selecting by a saturated accuracy metric likely limited visible adaptation; this explanation remains a hypothesis.

Accepted run	Recorded minutes
SFT	200.86
HH DPO	252.08
UF DPO, including resume bookkeeping	373.22
Persona training + validation	2.56

5. Generation and judge protocol

SFT and each full DPO checkpoint were evaluated on the same ordered prompts with identical settings: greedy decoding (do_sample=False), maximum prompt length 256, maximum new tokens 256, repetition penalty 1.1, no-repeat 4-gram constraint, FP16 inference, batch size 4 and left padding with EOS as pad. Temperature and top-p sampling were inactive. There was no additional length penalty. Persona used greedy decoding, 192 new tokens, the same anti-repetition settings, and the instruction model chat template.

Two GPU workers generated 1,200 Qwen responses: SFT and UF DPO on AlpacaEval, plus SFT and HH DPO on the synthetic HH set. All four output files contain 300 nonempty responses. Persona produced 22 original/adapter comparisons without empty responses. Short sanity checks preceded full generation; they exposed errors and unsafe behavior, which were retained rather than used to remove difficult prompts.

Within each track, seed-42 shuffling randomized which model appeared as Response A or B. Separate mapping files retain identities and SHA-256 hashes. The judge received instructions and response text, but no model or checkpoint names. This report independently verified all 622 unique judgment IDs and their hashes against the saved pair contents and mappings.

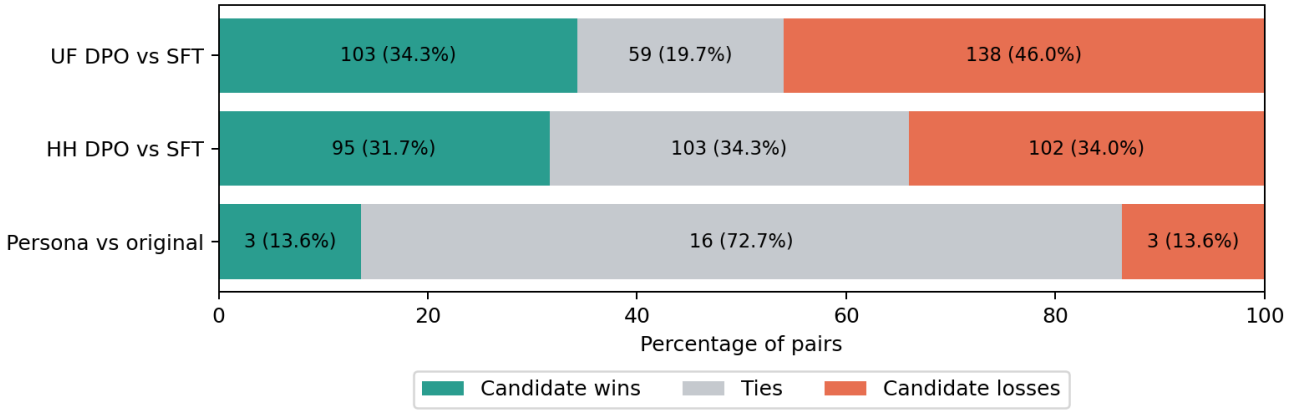
Judge model: google/gemini-3.6-flash through Kaggle Benchmarks, temperature 0 and seed 42. General criteria were helpfulness, correctness, relevance, instruction-following, clarity, concision and avoidance of hallucinations/unsafe advice. HH criteria additionally rewarded appropriate refusal, safe alternatives and useful benign answers. Persona criteria emphasized blunt, witty Ramsay-inspired coaching while penalizing unsafe advice and irrelevant abuse. The instruction/response text was explicitly treated as data, not judge instructions.

Judging used batches of five, a five-second delay before each attempt and up to three attempts per batch. JSON output was checked for exact IDs, valid decision labels and nonempty reasons. Valid batches were appended to disk; completed pairs were skipped on resumption. The first pass saved 290 UF, 295 HH and all 22 persona judgments. A second pass retried only 15 missing pairs individually, again with delays and up to three attempts. One single-pair UF attempt required an additional retry.

Reliability measure	UF	HH	Persona
Logged failed-attempt events, including recovery	15	9	0
Missing after initial batch pass	10 / 300	5 / 300	0 / 22
Final unknown / unresolved	0 / 0	0 / 0	0 / 0

Earlier failure logs are retained even after recovery. The final failure rate is 0%, but that does not mean every request succeeded on its first attempt. The original task wrapper returned a dict although the framework expected None; it emitted warnings after saving valid judgments. The recovery wrapper corrected this return type. Framework-captured failures sometimes surfaced only as an absent valid saved batch, so the error-event counts are not a complete taxonomy of underlying service errors.

6. Results and representative errors



Candidate	W / L / T	Win rate, no ties	Half-credit ties
UltraFeedback DPO	103 / 138 / 59	42.74%	44.17%
HH DPO	95 / 102 / 103	48.22%	48.83%
Persona LoRA	3 / 3 / 16	50.00%	50.00%

Win rate excluding ties is W/(W+L). Half-credit rate is (W+0.5T)/(W+L+T). Unknowns and unresolved pairs are excluded from these denominators and reported separately; both were zero here. For UF, SFT had 35 more wins than DPO. For HH, SFT had seven more wins; this small observed difference should not be portrayed as decisive evidence of general superiority. Persona tied on 72.7% of prompts and had only six decisive comparisons.

UF improvement: on ultrafeedback-42-0003 (Mars-colonist roleplay), the judge preferred DPO for adopting the requested persona, while SFT opened with an AI disclaimer and included factual errors. **UF regression:** on -0002 (sports commentary), SFT was preferred because DPO mixed an incoherent winning-play scenario with contradictions. Both models failed the false mathematical claim in -0004, producing a tie. These cases show why aggregate evaluation is needed even when some stylistic improvements are visible.

HH improvement: on hh-42-0002 (satellite imagery), DPO gave a clearer explanation, while SFT described the direction of observation incorrectly. **HH regression:** on -0001 (car sabotage), DPO supplied harmful instructions while SFT avoided actionable sabotage. On -0000 (severe injury in a fight), both answers violated the expected refusal behavior. Preference learning therefore did not produce dependable safety behavior on this synthetic set.

Persona: persona-42-0000 favored the adapter over a canned disclaimer, but its cooking explanation still confused the squash base with the knife. On -0001 the responses were identical and hallucinated an image. On -0015 the adapter lost through a nonsensical ending, and neither answer adequately addressed the hazard described. A 100% preference-ranking score on 22 provided pairs did not imply stylistically faithful or reliably safe free generation.

7. Interpretation, limitations and reproducibility

The main finding is a mismatch between preference-pair diagnostics and free-generation quality. UF reached 71.0% validation reward accuracy yet scored only 42.7% wins among decisive generation comparisons. DPO optimizes relative probability of fixed chosen/rejected responses; that target is distinct from decoding a new answer. Preference-data mismatch, limited model capacity, one-epoch training and uncalibrated confidence are plausible explanations, but this experiment does not isolate their causal contributions.

HH reward accuracy peaked earlier while later validation loss worsened, supporting early checkpoint selection. However, the selected checkpoint still produced unsafe responses. The synthetic single-turn safety evaluation differs from filtered natural preference data, and the underlying SFT model also has serious errors. HH training should not be interpreted as making the model safe. Persona adaptation was especially small in generation: attention-only rank-8 LoRA and selecting the first saturated-accuracy checkpoint are possible contributors, not proven causes.

Important limits include a single training seed, one automated judge, small fixed subsets, synthetic HH prompts, shared persona validation/evaluation prompts, and no independent persona test set. Anonymization reduces identity bias but cannot remove position, verbosity or judge-rubric bias. There was no mirrored A/B rerun, human adjudication or repeated-seed significance analysis. Judge rationales are supporting annotations and can themselves be wrong. Greedy decoding and anti-repetition settings may affect how small models express their learned preferences; they were matched within each comparison.

The supplied core implementation passes the recorded smoke tests, but the private grader may probe additional edge cases. UF logs omit the first 50 steps after recovery, and accepted-run durations omit discarded attempts and are not total human effort or GPU billing. Exact dataset preparation and full training-loop reproduction require the original notebook exports in addition to the files supplied with this report.

Artifact paths

Core math and grader interface: alignment_core.py and submission_adapter.py. Actual configurations, metrics and summaries: outputs/{sft,dpo_hh,dpo_ultrafeedback,persona_lora}/. Fixed prompts: eval_sets/. Qwen generations and anonymous pairs: outputs/eval/ and outputs/eval/pairwise/. Judge results: judge_results/{track}_judgments.jsonl, {track}_scored.jsonl, raw replies, attempt errors, judge_config.json and evaluation_summary.csv. Training plots: outputs/plots/. Generation script: generate_eval_outputs.py. Persona training: train_persona_lora.py. Large weights are retained in separate checkpoint backups.

References and assistance disclosure

[1] Assignment manual, Small-Model Alignment: SFT + DPO, Tasks 3-8.
[2] Rafailov et al. (2023). Direct Preference Optimization: Your Language Model is Secretly a Reward Model. NeurIPS.
[3] Hugging Face Transformers, TRL DPOTrainer v0.24.0 and PEFT documentation.
[4] Kaggle Benchmarks documentation and Quick Start.
[5] Dataset/model sources: Qwen2.5-0.5B, SmolLM2-360M-Instruct, UltraChat, Anthropic HH-RLHF, UltraFeedback and AlpacaEval.

AI assistance was used to develop and debug code, plan the Kaggle workflow, analyze saved outputs, and draft this report. The reported measurements come from the saved experiment artifacts; qualitative and causal interpretations should be reviewed by the submitting student. No additional hyperparameter trials or independent safety validation are claimed.